

Name: _____

Points: 16 /20

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Databasutveckling och lagring Final Exam

2025-02-19

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an X the right answer.
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

There are **16 questions**, each question gives **1 point**, except the last question which gives **5 points**; for a total of maximum **20 points**.

Grading

Points	Grade
0 - 9	IG
10 - 17	G
18 - 20	VG

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READ EACH QUESTION CAREFULLY!

1 SQLite and MariaDB are examples of

- 1. NoSQL databases
- ✓ 2. Relational Databases
- 3. In-memory databases

2 A table is

- ✓ 1. A collection of structured data
- 2. A technique used to normalize data
- 3. A type of data we can use in our databases

3 A column is

- ✓ 1. A technique to normalize data
- 2. A field with a certain datatype in a table
- 3. A type of data we can use in our databases

4 A Primary Key

- 1. Is more important than a secondary key
- ✓ 2. Is a deprecated technique no longer used in modern RDBMS
- 3. Identifies each row in a table uniquely

5 We keep the state of our data consistent at all times by using

- ✓ 1. Transactions
- 2. Transfusions
- 3. Transitions
- 4. Transmissions
- 5. Translations
- 6. Transformers

6 Given I have a table called users, and I want to set the field active to 0 for all users who have not confirmed their email address (stored in the boolean field email_confirmed), I can run

1. UPDATE * FROM users WHERE active = 0 AND email_confirmed = 0
2. UPDATE users WHERE active = 0 AND email_confirmed = 0
3. UPDATE * FROM users SET email_confirmed = 0 WHERE active = 0
4. UPDATE users SET email_confirmed = 0 WHERE active = 0
5. UPDATE * FROM users SET active = 0 WHERE email_confirmed = 0
6. UPDATE users SET active = 0 WHERE email_confirmed = 0

7 Database normalization allow us to

1. Save more data with less storage space
2. Retrieve more data with less memory usage
3. Retrieve more data with less bandwidth usage
4. Avoid anomalies
5. Avoid abnormalities

8 1NF states that

1. Each field in a table cannot contain multiple values
2. Each table cannot contain multiple fields
3. Each database cannot contain multiple tables
4. Each table must have a primary key - det at start.

9 To conform to 2NF, a table must conform to 1NF and

1. The primary key must depend on all non-key values
2. All non-key values must depend on the entire primary key

10 To conform to 3NF, a table must conform to 1NF

1. True

2. False

11 How can I take a backup of a SQLite database?

1. `sqlite3 database.db "-backup > backup.sql`
2. `sqlite3 database.db -dump > backup.sql`
- ✓ 3. `sqlite3 database.db .backup > backup.sql`
4. `sqlite3 database.db .dump > backup.sql`

12 How can I restore a SQLite backup?

1. `sqlite3 database.db | backup.sql`
- 2. `sqlite3 database.db < backup.sql`
3. `sqlite3 database.db -restore backup.sql`
- ✗ 4. `sqlite3 database.db .restore backup.sql`

13 Indices will speed up all operations on the table

1. True

✓ 2. False

14 How many days did Hanukkah of Data consist of?

1. 6

✓ 2. 7

3. 8

15 When using SQLite with python, by default INSERT and UPDATE operations need to be manually committed

✓ 1. True

2. False

16 Explain what a JOIN is (5 points)

✓ Join is when you can combine ^{values in} different tables with foreign key, you combine values.
